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Facultad de Ciencias – Departamento de Física

Soluciones a los problemas vistos en clases: 03-08 al 07-08

Álgebra de vectores: Actividades 1.3, 1.4 y 1.5

Actividad 1.3

Dados los vectores

$$\vec{A} = -3,45\hat{i} - 0,84\hat{j} + 1,21\hat{k}, \quad \vec{B} = (4\pi \times 10^{-1})\hat{i} + (0,04 \times 10^2)\hat{j} + 1,7\hat{k}, \quad \vec{C} = \hat{i} - 6\hat{j} - 8\hat{k}.$$

Primero conviene reducir las componentes de $\vec{B}$:

$$B_x = 4\pi \times 10^{-1} = 0,4\pi \approx 1,25664, \quad B_y = 0,04 \times 10^2 = 4, \quad B_z = 1,7,$$

$$\Rightarrow \vec{B} \approx 1,25664\hat{i} + 4\hat{j} + 1,7\hat{k}.$$

$$(1) \quad \left\| 2\vec{A} - \sqrt{3}\vec{B} + 0,361\vec{C} \right\|$$

Paso 1 — escalar cada vector:

$$2\vec{A} = (-6,90, -1,68, 2,42),$$

$$\sqrt{3}\vec{B} = 1,73205(1,25664, 4, 1,7) = (2,17656, 6,92820, 2,94449),$$

$$0,361\vec{C} = (0,361, -2,166, -2,888).$$

Paso 2 — combinar componente a componente ($2\vec{A} - \sqrt{3}\vec{B} + 0,361\vec{C}$):

$$x = -6,90 - 2,17656 + 0,361 = -8,71556,$$

$$y = -1,68 - 6,92820 - 2,166 = -10,77420,$$

$$z = 2,42 - 2,94449 - 2,888 = -3,41249.$$

$$\Rightarrow 2\vec{A} - \sqrt{3}\vec{B} + 0,361\vec{C} = -8,71556\hat{i} - 10,77420\hat{j} - 3,41249\hat{k}.$$



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Paso 3 — módulo:

$$\begin{aligned}\|\cdots\| &= \sqrt{(-8,71556)^2 + (-10,77420)^2 + (-3,41249)^2} \\ &= \sqrt{75,961 + 116,083 + 11,645} = \sqrt{203,689}.\end{aligned}$$

$$\left\| 2\vec{A} - \sqrt{3}\vec{B} + 0,361\vec{C} \right\| \approx 14,27$$

$$(2) \quad \frac{\vec{A}}{\sqrt{3}} - \left(\frac{\vec{B}}{2} + \frac{\vec{C}}{\sqrt{\sin 20^\circ}} \right)$$

Con $\sin 20^\circ = 0,34202$ (ángulo en grados), $\sqrt{\sin 20^\circ} = 0,58482$.

Paso 1 — escalar cada término:

$$\begin{aligned}\frac{\vec{A}}{\sqrt{3}} &= \frac{1}{1,73205}(-3,45, -0,84, 1,21) = (-1,99186, -0,48497, 0,69859), \\ \frac{\vec{B}}{2} &= (0,62832, 2, 0,85), \\ \frac{\vec{C}}{\sqrt{\sin 20^\circ}} &= \frac{1}{0,58482}(1, -6, -8) = (1,70991, -10,25947, -13,67929).\end{aligned}$$

Paso 2 — sumar el paréntesis $\left(\frac{\vec{B}}{2} + \frac{\vec{C}}{\sqrt{\sin 20^\circ}} \right)$:

$$(0,62832 + 1,70991, 2 - 10,25947, 0,85 - 13,67929) = (2,33823, -8,25947, -12,82929).$$

Paso 3 — restar del primer término:

$$\begin{aligned}x &= -1,99186 - 2,33823 = -4,33009, \\ y &= -0,48497 - (-8,25947) = 7,77450, \\ z &= 0,69859 - (-12,82929) = 13,52788.\end{aligned}$$

$$\frac{\vec{A}}{\sqrt{3}} - \left(\frac{\vec{B}}{2} + \frac{\vec{C}}{\sqrt{\sin 20^\circ}} \right) \approx -4,33\hat{i} + 7,77\hat{j} + 13,53\hat{k}$$

(su magnitud es $\sqrt{4,33009^2 + 7,77450^2 + 13,52788^2} = \sqrt{262,20} \approx 16,19$).

Actividad 1.4

A partir de la figura (cubo de aristas 5, 10 y 8 [m]) se leen los vectores

$$\vec{A} = (10, 8, -5), \quad \vec{B} = (-8, 10, -5), \quad \vec{C} = (8, -10, 0).$$

(1) $-3\vec{A} + 2(\vec{C} \times \vec{B})$

Paso 1 — producto cruz $\vec{C} \times \vec{B}$ (determinante y expansión por cofactores):

$$\begin{aligned}\vec{C} \times \vec{B} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & -10 & 0 \\ -8 & 10 & -5 \end{vmatrix} \\ &= \hat{i}[(-10)(-5) - (0)(10)] - \hat{j}[(8)(-5) - (0)(-8)] + \hat{k}[(8)(10) - (-10)(-8)] \\ &= \hat{i}[50] - \hat{j}[-40] + \hat{k}[80 - 80] = (50, 40, 0).\end{aligned}$$

Paso 2 — escalar y sumar:

$$-3\vec{A} = (-30, -24, 15), \quad 2(\vec{C} \times \vec{B}) = (100, 80, 0),$$

$$-3\vec{A} + 2(\vec{C} \times \vec{B}) = (-30 + 100, -24 + 80, 15 + 0).$$

$$\boxed{-3\vec{A} + 2(\vec{C} \times \vec{B}) = 70\hat{i} + 56\hat{j} + 15\hat{k}}$$

(2) $(\vec{A} \times \vec{B}) - (\vec{C} \times \vec{A})$

Paso 1 — $\vec{A} \times \vec{B}$:

$$\begin{aligned}\vec{A} \times \vec{B} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10 & 8 & -5 \\ -8 & 10 & -5 \end{vmatrix} \\ &= \hat{i}[(8)(-5) - (-5)(10)] - \hat{j}[(10)(-5) - (-5)(-8)] + \hat{k}[(10)(10) - (8)(-8)] \\ &= \hat{i}[-40 + 50] - \hat{j}[-50 - 40] + \hat{k}[100 + 64] = (10, 90, 164).\end{aligned}$$

Paso 2 — $\vec{C} \times \vec{A}$:

$$\begin{aligned}\vec{C} \times \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & -10 & 0 \\ 10 & 8 & -5 \end{vmatrix} \\ &= \hat{i}[(-10)(-5) - (0)(8)] - \hat{j}[(8)(-5) - (0)(10)] + \hat{k}[(8)(8) - (-10)(10)] \\ &= \hat{i}[50] - \hat{j}[-40] + \hat{k}[64 + 100] = (50, 40, 164).\end{aligned}$$

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Paso 3 — restar:

$$(\vec{A} \times \vec{B}) - (\vec{C} \times \vec{A}) = (10 - 50, 90 - 40, 164 - 164).$$

$$\boxed{(\vec{A} \times \vec{B}) - (\vec{C} \times \vec{A}) = -40\hat{i} + 50\hat{j} + 0\hat{k}}$$



Actividad 1.5

Dados los vectores

$$\vec{A} = -9\hat{i} + 12\hat{j} - 8\hat{k}, \quad \vec{B} = 4\hat{i} - 5\hat{j} + 10\hat{k}.$$

(1) Producto escalar $\vec{A} \cdot \vec{B}$

$$\vec{A} \cdot \vec{B} = (-9)(4) + (12)(-5) + (-8)(10) = -36 - 60 - 80.$$

$$\boxed{\vec{A} \cdot \vec{B} = -176}$$

(2) Producto vectorial $\vec{A} \times \vec{B}$

$$\begin{aligned} \vec{A} \times \vec{B} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -9 & 12 & -8 \\ 4 & -5 & 10 \end{vmatrix} \\ &= \hat{i}[(12)(10) - (-8)(-5)] - \hat{j}[(-9)(10) - (-8)(4)] + \hat{k}[(-9)(-5) - (12)(4)] \\ &= \hat{i}[120 - 40] - \hat{j}[-90 + 32] + \hat{k}[45 - 48] = (80, 58, -3). \end{aligned}$$

$$\boxed{\vec{A} \times \vec{B} = 80\hat{i} + 58\hat{j} - 3\hat{k}}$$

(3) Ángulo entre $\vec{A}$ y $\vec{B}$ (usando el producto punto)

Paso 1 — módulos:

$$|\vec{A}| = \sqrt{(-9)^2 + 12^2 + (-8)^2} = \sqrt{81 + 144 + 64} = \sqrt{289} = 17,$$

$$|\vec{B}| = \sqrt{4^2 + (-5)^2 + 10^2} = \sqrt{16 + 25 + 100} = \sqrt{141} \approx 11,8743.$$

Paso 2 — despejar el ángulo de $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos\theta$:

$$\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} = \frac{-176}{17\sqrt{141}} = \frac{-176}{201,864} = -0,87188.$$

$$\boxed{\theta = \arccos(-0,87188) \approx 150,7^\circ}$$



Problema 1

Dados los vectores

$$\vec{A} = -0,05\hat{i} + 0,08\hat{j} - 0,10\hat{k}, \quad \vec{B} = 0,25\hat{i} + 0\hat{j} - 0,10\hat{k},$$

determine:

(1) $\frac{1}{3}(\vec{A} \cdot \vec{B})$

$$\vec{A} \cdot \vec{B} = (-0,05)(0,25) + (0,08)(0) + (-0,10)(-0,10) = -0,0125 + 0 + 0,01 = -0,0025.$$

$$\boxed{\frac{1}{3}(\vec{A} \cdot \vec{B}) = \frac{-0,0025}{3} \approx -8,33 \times 10^{-4}}$$

(2) $\frac{1}{5}(\vec{B} \times \vec{A})$

$$\begin{aligned} \vec{B} \times \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0,25 & 0 & -0,10 \\ -0,05 & 0,08 & -0,10 \end{vmatrix} \\ &= \hat{i}[(0)(-0,10) - (-0,10)(0,08)] - \hat{j}[(0,25)(-0,10) - (-0,10)(-0,05)] + \hat{k}[(0,25)(0,08) - (0)(-0,05)] \\ &= \hat{i}[0,008] - \hat{j}[-0,025 - 0,005] + \hat{k}[0,02] = (0,008, 0,030, 0,02). \end{aligned}$$

$$\boxed{\frac{1}{5}(\vec{B} \times \vec{A}) = (0,0016, 0,006, 0,004) = 1,6 \times 10^{-3}\hat{i} + 6 \times 10^{-3}\hat{j} + 4 \times 10^{-3}\hat{k}}$$

(3) **Ángulo entre $\vec{A}$ y $\vec{B}$**

$$|\vec{A}| = \sqrt{0,05^2 + 0,08^2 + 0,10^2} = \sqrt{0,0189} \approx 0,13748, \quad |\vec{B}| = \sqrt{0,25^2 + 0,10^2} = \sqrt{0,0725} \approx 0,26926.$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \frac{-0,0025}{(0,13748)(0,26926)} = \frac{-0,0025}{0,037018} \approx -0,06754.$$

$$\boxed{\theta = \arccos(-0,06754) \approx 93,87^\circ}$$



Problema 2

Para el paralelepípedo rectangular de lados 2, 4 y 6 [m] (Figura 1), las coordenadas cartesianas de los vectores son

$$\vec{A} = (2, 6, 0), \quad \vec{B} = (0, -6, 0), \quad \vec{C} = (2, -6, 4).$$

$$(1) \quad \frac{\vec{A} - \vec{B}}{\|\vec{A} + \vec{B}\|^3}$$

$$\begin{aligned} \vec{A} - \vec{B} &= (2, 12, 0), & \vec{A} + \vec{B} &= (2, 0, 0), \\ \|\vec{A} + \vec{B}\| &= \sqrt{2^2 + 0^2 + 0^2} = 2 \Rightarrow \|\vec{A} + \vec{B}\|^3 = 8. \end{aligned}$$

$$\frac{\vec{A} - \vec{B}}{\|\vec{A} + \vec{B}\|^3} = \frac{(2, 12, 0)}{8} = 0,25\hat{i} + 1,5\hat{j}$$

$$(2) \quad \frac{\vec{C} + 2\vec{A}}{\|\vec{B}\|^2}$$

$$\begin{aligned} 2\vec{A} &= (4, 12, 0), & \vec{C} + 2\vec{A} &= (6, 6, 4), \\ \|\vec{B}\| &= \sqrt{0^2 + (-6)^2 + 0^2} = 6 \Rightarrow \|\vec{B}\|^2 = 36. \end{aligned}$$

$$\frac{\vec{C} + 2\vec{A}}{\|\vec{B}\|^2} = \frac{(6, 6, 4)}{36} = \frac{1}{6}\hat{i} + \frac{1}{6}\hat{j} + \frac{1}{9}\hat{k} \approx (0,167, 0,167, 0,111)$$

$$(3) \quad \vec{C} \cdot \vec{B}$$

$$\vec{C} \cdot \vec{B} = (2)(0) + (-6)(-6) + (4)(0) = 0 + 36 + 0.$$

$$\vec{C} \cdot \vec{B} = 36$$



$$(4) \quad \vec{B} \times 2\vec{A}$$

$$\begin{aligned} \vec{B} \times 2\vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -6 & 0 \\ 4 & 12 & 0 \end{vmatrix} \\ &= \hat{i}[(-6)(0) - (0)(12)] - \hat{j}[(0)(0) - (0)(4)] + \hat{k}[(0)(12) - (-6)(4)] \\ &= \hat{i}[0] - \hat{j}[0] + \hat{k}[24]. \end{aligned}$$

$$\boxed{\vec{B} \times 2\vec{A} = 24\hat{k}}$$

